(Please write your Enrollment Number) Enrollment No._____

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B. Tech, Semester: 5th
(December, 2025)

Subject Code: BAI 303	Subject: Cyber Security
Time :3 Hours	Maximum Marks: 60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(4*5=20)	CO Mapping
a)	Discuss the process of ARP spoofing. Explain how ARP cache poisoning occurs and the impact it has on data confidentiality and integrity.	(04)	CO1
b)	Describe the concept of a Reflection Attack. How does IP spoofing enable reflection-based DDoS?	(04)	CO1
c)	Explain the complete sequence of steps involved in a large-scale financial cyber fraud committed through compromised digital payment identities.	(04)	CO2
d)	Compare symmetric and asymmetric cryptographic systems.	(04)	CO3
e)	Compare the Lollipop Model and the Onion Model of defense in terms of structure, assumptions, and practical security implications.	(04)	CO4
	UNIT I		CO Mapping
Q2	The Dark Web supports illegal marketplaces and cybercrime ecosystems. Explain the technical architecture and operational workflow of the following: • Tor-based anonymity • Onion routing process • Role of exit nodes • How cybercriminals use these components for APTs, ransomware operations, and data exfiltration.	(10)	CO1
Q3	A multinational company faces repeated phishing incidents leading to credential theft. Analyze this case by explaining: • The phishing attack chain • How attackers craft convincing baits • Credential harvesting mechanisms • Browser-level vulnerabilities exploited • Long-term defense processes for: users, browsers, networks, and servers.	(10)	CO1
	UNIT - II		CO Mapping
Q4	Discuss the socio-technical consequences of security breaches on social networking websites, addressing individual and organizational dimensions. Give threat matrix for the same explaining different kind of attacks their impact and counter measure.	(10)	CO2
Q5	Discuss the personal and organizational impacts of security breaches in smartphones and smart devices. Present a threat matrix outlining major attacks, their effects, and corresponding countermeasures.	(10)	CO2

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	UNIT III		CO Mapping
Q6	A toy hash function processes data in **8-bit blocks** using the rule: $$H_i = (H_{i-1} \oplus B_i) \oplus R$$ Where: • $H_0 = 11110000$(initial value) • Round constant $R = 00111100$ • Message blocks: ○ $B_1 = 10101010$ ○ $B_2 = 11001100$ a) Compute H_1 b) Compute H_2 c) State one limitation of simple XOR-based hash constructions.	(10)	CO3
Q7	Compare SSL/TLS, IPSec, and SSH as secure communication protocols, focusing on their design goals, layer of operation, and practical deployment contexts.	(10)	CO3
	UNIT IV		CO Mapping
Q8	Explain the purpose of access control in organizational security. Evaluate the role of Network Admission Control in enforcing endpoint compliance within enterprise networks, emphasizing its contribution to organizational defense.	(10)	CO4
Q9	Define intrusion detection. Contrast ISO/IEC 27001 and the NIST Cybersecurity Framework in terms of scope, implementation approach, and organizational applicability.	(10)	CO4

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